

Contractils: Quantum Mechanical Fibrils

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ABSTRACT: When an erythrocyte rouleau is pulled apart by micro-manipulation the cells remain attached to each other by long fibrils which may reach lengths of over 100 μm . The tension and length of the fibrils are drastically reduced when the cellular membrane potential is abolished or when the cells are depleted of ATP. It is inferred that the fibrils consist of macromolecules and that the tension is largely derived from a quantum mechanical long-range interaction which accords with the theory of Fröhlich.

INTRODUCTION

HERBERT FRÖHLICH'S THEORY (1968, 1980) of a phonon-maser-like phenomenon in cell membranes predicts intercellular forces of range several orders of magnitude longer than London-van der Waals' force. Interactions between human erythrocytes at a range of several micrometers have recently been demonstrated by deviation from Brownian movement (Rowlands et al. 1981, 1982a, 1982b). Essential to Fröhlich's hypothesis are (a) an organized array of molecules in a membrane, (b) a membrane potential to polarize membrane molecules and (c) a supply of metabolic energy which, if it exceeds a minimum value, will trigger membrane vibrations into a coherent polar oscillatory mode ($\nu \approx 10^{11}\text{Hz}$), a process analogous with the transition to coherence in a maser or in a laser. When the membrane of the erythrocyte is disorganized with glutaraldehyde the deviations from Brownian movement disappear and they disappear *reversibly* when the membrane potential is reduced to zero or if the cells are deprived of ATP (Rowlands et al. 1981, 1982a, 1982b). The presence of macromolecules in the medium suspending the erythrocytes is essential to the interaction (Rowlands et al. 1982b; Sewchand et al. 1982b) and although the necessary and sufficient properties of the macromolecules have yet to be delineated they need to have an extended chain structure.

In the experiments described by Rowlands et al. the Fröhlich interaction is superimposed on Brownian movement and the two motions are comparable in magnitude. When erythrocytes in Brownian motion collide they adhere and quite quickly (in the time frame of the experiment) slide face to face. Further cells and groups of cells adhere and so columns of erythrocytes develop, which is the well-known process of erythrocyte rouleau formation. In the present work rouleaux are pulled apart by micromanipulation. In the process long contractile fibrils (contractils) form between the cells. The properties of these contractils vary when membrane function is interfered with by the methods (a), (b) and (c) above and these variations provide additional evidence for a Fröhlich mechanism in human erythrocytes.

METHODS

Fresh heparinized blood with the haematocrit adjusted to about 1% was placed on a large cover slip on an inverted microscope (Leitz Diavert, Rockleigh, NJ). In some experiments a tiny chamber was constructed on the cover slip with

pieces of microscope slides and of cover slips. Micropipettes could be inserted into a specimen of blood in the chamber and then the whole preparation sealed with medicinal paraffin. These latter experiments showed that the observed phenomena were not a consequence of the specimen drying out. In both types of experiment the end cells of a rouleau were held by gentle suction on each of two diametrically opposed micropipettes of internal diameter $<3\ \mu\text{m}$. Canham and Burton (1968) showed that human erythrocytes have a "minimum cylindrical diameter" of $3\ \mu\text{m}$. That is, a cell will deform into a cylinder, with hemispherical ends, of this diameter without change of volume or of surface area. With a pipette of internal diameter $<3\ \mu\text{m}$ gentle suction pulls part of the cell into the pipette and thus the cell is firmly held. If the pipettes are now separated, so as to exert traction on the rouleau, the cells do not fall apart but remain linked by fibrils of length $15\ \mu\text{m}$ or more (Fig. 1). If the negative pressure in one pipette is cut off, or if a fibril breaks, the remaining fibrils contract, pulling the cells through the plasma at velocities of the order of tens of micrometers per second. Observations were recorded with a still camera, a television camera, and tape recorder, or with a 16mm. cine-camera.

Experimental modification of the cells. The methods used were those fully described in Rowlands et al. (1982b). In brief, the membrane potential was brought close to zero by reducing the pH of plasma to 6.3 and restored by returning it to pH = 7.4. (Funder and Wieth, 1966; Lassen et al. 1980). The membrane potential was calculated from measurements of the chloride ratio $[\text{Cl}_i^-/\text{Cl}_o^-]$ using the Nernst equation. The other (nonreversible) method of reducing the membrane potential involved the use of the ionophore A23187 and the procedure of Flatman and Lew (1980). Metabolic depletion was achieved by incubation for 24 hr at 37°C and cells were repleted by further incubation after adding 30 mM adenosine (Meiselman and Baker 1977). The membrane was disorganized by a 24 hr incubation in 0.04% glutaraldehyde (Rowlands et al. 1982b). In whatever manner the cells were modified, they were resuspended in their own plasma, except in the following experiments.

Mixed species experiments. Rouleaux composed of a mixture of cells from man and rabbit were examined. The cells cannot be suspended in human plasma because of immunological reactions. The suspending liquid was 0.4g% polyvinylpyrrolidone (molecular weight 360,000) in standard phosphate-buffered saline. Human serum albumin (0.2g%) was added to prevent crenation of the cells. Polyvinylpyrrolidone is a rouleau-inducing agent and it also transmits the long-range Fröhlich interaction between human erythrocytes (Sewchand et al. 1982b).

Scanning Electron Microscopy. The contractils are very thin and we have not succeeded, with the light microscope, in making a picture which demonstrates contractils unequivocally. The two younger authors of this paper see contractils by direct vision, but the old one never does. We therefore

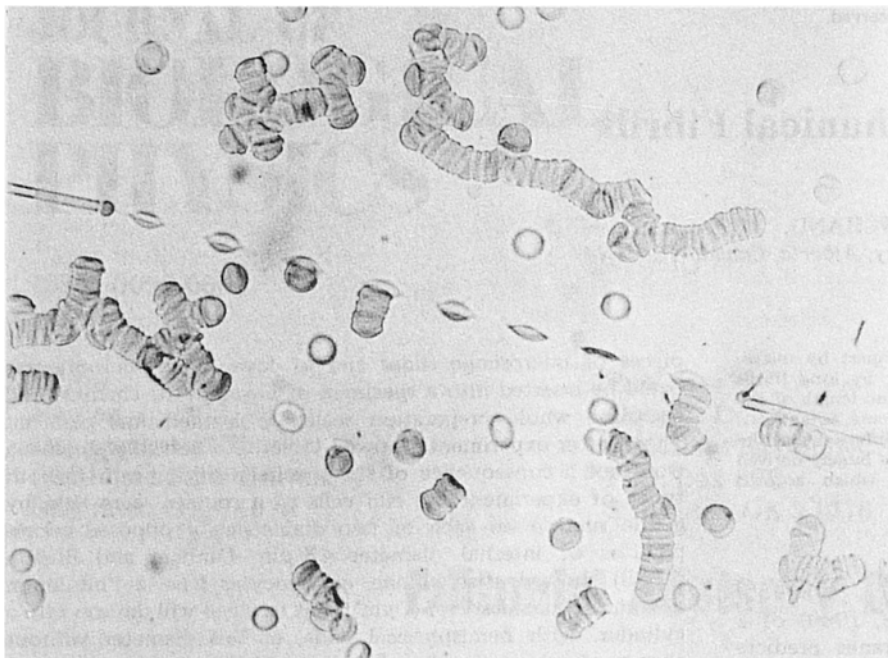


Fig. 1 A branched rouleau has been extended between the micropipettes seen at extreme left and right. The internal diameter of the pipettes is less than the cellular minimum cylindrical diameter (see text) and so the end cells are firmly held by negative pressure. Branches of the original rouleau are seen still attached to cells three, four, and eight of the chain. Human erythrocytes in their own plasma. Red cell diameter $\approx 8 \mu\text{m}$.

embarked on scanning electron microscopy. In the end success was achieved but the technique demands the utmost delicacy. Contractils were pulled in an open chamber which could be perfused. With a satisfactory chain of cells in view through the light microscope, glutaraldehyde (0.5g% in phosphate-buffered saline) slowly replaced the rouleau-inducing fluid. This was followed by flow of distilled water (40 ml) to remove all the glutaraldehyde solution. The water was sucked out of the chamber and the specimen was allowed to dry overnight, still on the light microscope with micromanipulators and pipettes in place. If the specimen looked satisfactory the next day, the micropipettes were rigidly glued to the floor of the chamber with a drop or two of epoxy cement. When the cement was hard the outer parts of the pipettes were sheared off with heavy metal cutters. The flow chamber could now be taken from the light microscope. The glass floor of the chamber was removed and cut around the tips of the attached pipettes to a size to fit a metal sample holder. The rest was the standard technique of scanning electron microscopy. The thickness of the gold coating, according to the machine calibration, was 80 nanometres.

RESULTS

The chain of cells in Fig. 1 was drawn from a large branched rouleau such as seen elsewhere in the field of view. That is why small rouleaux are attached to some of the cells in the chain. The gaps in the chain are on the average $10 \mu\text{m}$ long. These are by no means the longest contractils we have seen. The maximum length of a contractil before breaking varies from rouleau to rouleau and from day to day in the blood of the same donor.

Glutaraldehyde-fixed cells. Our method leaves the shape, area, and volume of the cells unaffected. Such cells are rigid and cannot be drawn into a pipette. Rouleaux do not form. Occasionally cells will stick together and they can be separated by shear from the tips of the pipettes. There is no sign of a contractil. Once separated the cells fall apart.

Ionophore-treated (A23187) cells. Two experiments were done with identical results. Rouleaux formed without delay and their appearance was normal; Yet no contractils were

seen. As soon as the cell at one end of a rouleau was sucked into the first pipette, the cell came away. The cells could be picked off one by one from a rouleau with a single pipette.

Variation of membrane potential by varying pH (four experiments). At pH = 6.3 when the membrane potential is within 1.5 mV of zero (Rowlands et al. 1982b) some rouleaux fell apart on manipulation but others formed short ($<4 \mu\text{m}$) contractils which gave the appearance of being weak, perhaps because further extension led to rupture. No long contractils were formed. Restoring the pH to 7.4 gave contractils with most rouleaux and they were consistently longer ($>6 \mu\text{m}$) than at pH = 6.3.

Metabolic depletion and repletion (four experiments). The majority of metabolically depleted cells are crenated. Crenated cells do not form rouleaux but they often stick together. They can be sucked easily into a micropipette and on manipulation adhering cells immediately fall apart with no sign of a contractil. A minority of cells are normal in appearance and some of these form rouleaux. Some of the rouleaux fall apart on micromanipulation: others demonstrated contractils but always they were short ($<4 \mu\text{m}$) as in the pH = 6.3 low membrane potential experiments. When the cells were repleted with adenosine crenation disappeared and rouleaux formed normally. Contractils reappeared and they were $>8 \mu\text{m}$ in length but, as when the membrane potential was restored, fully normal contractils were rarely seen.

Interspecies Experiments. Rouleaux which form from mixtures of erythrocytes from different species of mammals are not random in their grouping of cells. (Sewchand and Canham 1976). This non-randomness is further evidence for a Fröhlich interaction between erythrocytes (Rowlands et al. 1982a,b). A human-rabbit mixed rouleau, comprises short runs of cells of the two species and the cells can be identified by size. On micromanipulation such a mixed rouleau always separates at a junction between a rabbit cell and a human cell. There is sometimes a short contractil at the break. But when a mixed rouleau is stripped until the remaining rouleau is all rabbit or all human and then pulled apart, normal contractils occur.

Scanning Electron Microscopy. In dozens of attempts a break always occurred in all but the shortest contractils subsequent to removal from the light microscope. The contractil shown in Fig. 2 is no exception. Such breaks can occur in any

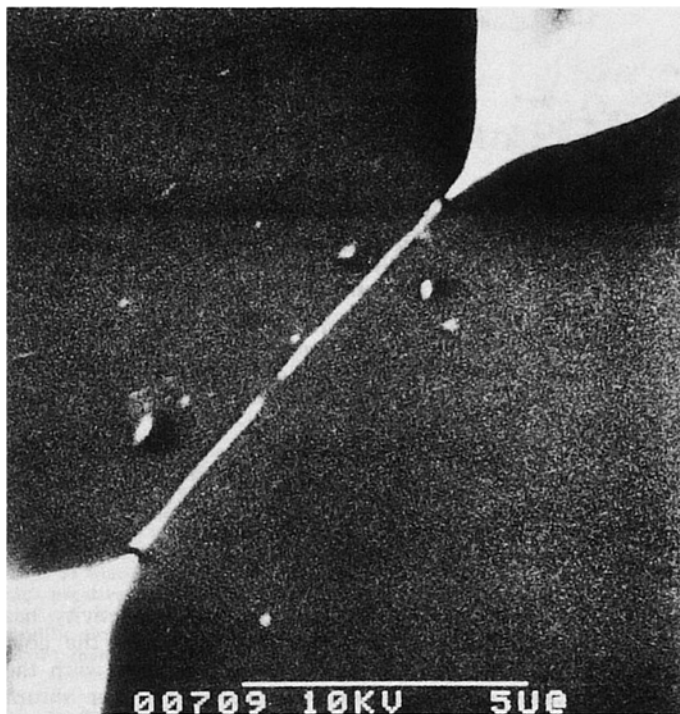


Fig. 2 Scanning electron microscope picture. See text for technique. The break in the middle of the contractil is an artefact. The breaks where the contractils begin to widen and merge with the contours of the cells are always present. The line above the identification marks is 5 μm long.

position along the fibre. Quite consistently, however, two additional and very short discontinuities are seen at the ends of each contractil just as it begins to widen and merge with the contours of the two cells to which it is attached. Since contractils and cells, after fixation, were washed in distilled water and then dried, the contractils as prepared for scanning electron microscopy, are stretched in air. Two breaks would cause them to fall and be displaced from alignment with the pointed aspects of the cells. Such displacement is never seen. This finding is considered in the Discussion.

Velocity of recoil. If a contractil breaks or if the negative pressure in one micropipette is released the stretched chain of cells recoils back into a rouleau. The maximum velocity of recoil measured by frame analysis of a cine-film was 180 $\mu\text{m}\cdot\text{s}^{-1}$. The length of the contractil at this velocity was 18 μm . By observing Brownian movement of erythrocytes resting on horizontal surfaces we have made extensive measurements on drag coefficients (Rowlands et al. 1982a,b; Sewchand et al. 1982a). If we assume for the purposes of an order of magnitude calculation that the cell was traveling at its terminal velocity under the force (F) from the contractil then

$$F = v\mu$$

where v is the velocity and μ is the drag coefficient. Such a calculation gives a value of the force of 10^{-6} N. If, from the width of the image of a contractil by scanning electron microscopy is subtracted the thickness of the gold shadow, the contractil itself comes out to be of width 0.2 μm (which explains the difficulty of resolution by light microscopy). From this a very rough estimate of the cross-sectional area and so of the stress in a contractil can be calculated giving 10^7 N/m². This must be regarded as an upper limit.

DISCUSSION

Coherent quantum mechanical excitations in cell mem-

branes as postulated by Fröhlich provide a mechanism for many cell processes which await explanation (Fröhlich 1977; Cooper 1979). There is evidence, from Raman spectroscopy, of vibrations of the predicted frequency. Irradiation with microwaves of microorganisms and of cells in tissue culture demonstrates resonances as frequency is varied. Fröhlich (1980) has reviewed this evidence. But the important prediction of the theory is for the existence of ultra-long-range specific interactions for which evidence is only recently beginning to accumulate (Pohl 1980,1981a,b,c; Rowlands et al. 1981, 1982a,b; Sewchand et al. 1982b). In brief Pohl has shown that particles of high dielectric constant will adhere to cells more readily than particles of low dielectric constant. Rowlands et al. have demonstrated an interaction between erythrocytes in their own plasma at a range of several micrometers. The interaction requires a membrane potential and a supply of energy.

The Rowlands et al. interaction takes place *before* cellular contact. Contractils arise *after* contact when the cells are pulled apart by micromanipulation. Certain macromolecules are essential to both processes and the inference is that the Fröhlich interaction of erythrocytes is transmitted by these macromolecules (Rowlands et al. 1982b; Sewchand et al. 1982b). Solutions of macromolecules can be pulled out into thin threads (e.g. polyethylene oxide) and it is no surprise that erythrocytes, adhering by virtue of suspension in a macromolecular solution, maintain adherence by thin threads when pulled apart. It is therefore necessary to review the evidence that the threads are powered and that the power is compatible with the predictions of Fröhlich's theory.

Glutaraldehyde-fixed cells. Although suspended in the same medium, fixed cells do not form contractils. They are, however, much more rigid than normal cells and so their area of contact is small. Therefore the area from which macromolecules could be recruited into a thread is much smaller than for normal cells. Thus the absence of contractils with fixed cells might merely be an area effect.

Ionophore-treated cells. These form normal-looking rouleaux which fall apart on micromanipulation. They are not so flexible as normal cells but they have a much larger area of contact than glutaraldehyde-fixed cells, as large as for normal cells. But reduced flexibility could prevent the macromolecules holding the cells together from being drawn towards the final point of membrane contact as the cells come apart.

Change of membrane potential by change in pH. At pH = 6.3 when the membrane potential was close to zero, rouleaux occur and sometimes these formed short, weak threads. The cells were of normal flexibility, the area of contact was maximal, and so these threads were probably a passive drawing out of the macromolecules which had held the membranes together. They never exceeded 4 μm in length and they broke within a few seconds. With normal cells the threads are very much longer, at least 10 μm and sometimes over 100 μm in length. Moreover threads from normal cells continue to exist for half an hour or more prior to fixation in the scanning electron microscope experiments. Restoring pH to 7.4 and so the membrane potential nearly to normal (Rowlands et al. 1982b) partially restored the contractils. They were longer and longer lasting than at pH = 6.3 but we never observed the very long contractils seen with normal cells.

Metabolic depletion and repletion. No contractils formed when the depleted cells were crenated and they reappeared when the metabolism was restored with adenosine. A minority of depleted cells have normal appearance, but formed rouleaux and weak contractils. These cells may not have been depleted as much as those which crenated.

Interspecies experiments. The nonrandomness of the arrangement of the cells in mixed rouleaux has been adduced as



Fig. 3 The pipettes are offset and the axis of the contractils is seen to be radial with respect to the spherical external part of each of the two cells held in the micropipettes. As the extent of the offset is varied the contractils always point towards the centres of the spheres. Human erythrocytes in their own plasma. Red cell diameter $\approx 8 \mu\text{m}$.

evidence for specificity in the interaction of erythrocytes (Rowlands et al. 1982a,b). The present experiments clearly demonstrate an adhesiveness between cells of different species which is lower than that between cells of the same species.

General Discussion. The experiments described in this paper are technically difficult and the results are qualitative. Partly this is a consequence of the technique but mainly because there is considerable variation in the properties of the cells in a single sample from an individual. There is for instance a very wide variation in the permeability of single cells to various non-electrolytes (Jay and Rowlands 1975). One "defective" cell in a rouleau would vitiate contractil formation. Such defective cells might be ripe for destruction in the spleen. Nevertheless the properties of these threads vary regularly in the manner expected if their tension is derived, in the main, from the living state of the erythrocytes. There are two further important pieces of evidence. When one contractil breaks the remaining cells are pulled together at great speed by the remaining contractils back into a rouleau. If the thread was just visco-elastic, it would shorten, but not down to a zero length, so rapidly bringing the membranes again into contact. Moreover a passive thread would always attach at the point on the membrane from which it originated. Reference to Fig. 3 shows that the line of attachment of the contractil to the spherical portion of the end cells held in the micropipettes is radial, directed towards the center of the sphere. As the pipettes are moved out of and into alignment, with the tips kept at a fixed distance apart, the attachment moves over the spherical surfaces, always directed towards the center of the sphere without hysteresis. If the pressure in the pipette is released slightly so that the sphere sags, then a point appears at the attachment which also moves freely with the contractil as its orientation is changed.

We conclude that the observed fibrils have an active contraction powered by the cells and that the contraction is consistent with the interaction postulated by Fröhlich.

An unequivocal picture of a contractil by light microscopy was not obtained. What two out of three of the authors saw might have been an optical illusion because the line of a contractil is clearly oriented by the direction of the points on the parent cells. Scanning electron microscopy proved the existence of a physical thread, which is composed of macromolecules. The constant gaps in the contractils which occur just as they merge into the cell outline are difficult to explain. By the time the gold, which is what is seen, falls on the contractils they and the cells have been fixed in glutaraldehyde, washed in distilled

water and air dried overnight, so all quantum activity has ceased. There must still be continuity otherwise the gold threads would have been displaced from alignment with the points of the cells. Clearly there must be a rather abrupt thinning of the contractil close to the pointed distortion of the cell membrane. Rowlands et al. (1982b) have suggested that the interaction is transmitted by quantum quasi-particles which may be tunnelling at this point between the membrane and the contractil.

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